

BINLEVEL · CALCULATION

BinLevel sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	First issue for TRL 3 (measurement geometry, accuracy, energy, supply, airtime and payload, link budget, timing, lid temperature, mass, shock, cost)

BinLevel sizing calculations

On paper, BinLevel meets eight of its sixteen requirements (four by calculation and four by design), has three at risk, misses three and leaves two that only a test can settle. The three misses are R5 (the stock enclosure is IP67, not IP69K), R10 (an antenna inside a steel container loses the link beyond about 0.35 km) and R16 (the unit weighs about 435 g against 400 g, because the stainless bracket alone is 183 g). The three at risk are accuracy on real waste (R2), the temperature range (R6, since a dark lid reaches about 67 °C in hot sun) and the temperature alert (R9, whose rate-of-rise trigger a cloud clearing could set off). Battery life is not a constraint: a C cell lasts about 16 years in the worst radio case. The calculations changed four details of the TRL 2 concept: the payload shrinks from 12 to 11 bytes so that it fits the slowest US915 data rate, the carrier gains a nanopower regulator because a fresh cell exceeds the module's 3.6 V limit, the temperature is read every 5 min so that the heat alert arrives within 15 min, and the cell is strapped rather than held by clips alone. Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [C3], is the line of that script's output that carries it.

Safety: These are first-principles estimates for a paper proof of concept. The temperature alert is a maintenance aid and not fire detection. Nothing here replaces checks of the lithium cell's fusing and retention, or safe working practice around bins and collection vehicles. See BNL-PRC-001, Safety.

■ Scope and method

The note checks every requirement in BNL-REQ-001 v0.3 against the design in BNL-PRC-001 v0.3 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS`, `derived()` and part solids, so the container depth, sensor positions, bracket volume and envelope used here are the ones in the STEP files and in drawing BNL-DWG-001. The script also reads `bom/bom.csv` and `budget_usd` in `project.yaml`. Run it from the repo root with `python docs/04-calcs/sizing.py`.

The design case is an 1,100 L four-wheel communal container (DDR-001, D1) with the sensor at the center of an HDPE lid, ranging every 15 min, reporting hourly (2 h at SF11 and SF12) and on alerts (D6), on a LoRaWAN network with a TwinKit or public gateway (D4, D5).

■ Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
Container	Body 1,300 x 1,000 mm outside, 8 mm wall, inner floor 228 mm and rim 1,300 mm above ground; lid wall 5 mm HDPE	Typical EN 840 1,100 L size (estimate), as in the model
Sensing	Ultrasonic blind zone 0.25 m and range to 4 m; effective beam half-angle 15°; ToF used to 0.4 m with a 27° field of view and ± 20 mm error including the window	JSN-SR04T-class figures are estimates to confirm; the VL53L1X field of view and 4 m maximum are from the ST product page
Accuracy	Air temperature known to ± 5 K from the on-board sensor; echo detection jitter of half a wavelength; 1 μ s timer	Assumed
Module	RAK3172: 2.0 to 3.6 V supply, 1.69 μ A sleep, -20 to 85 °C (the -T variant -40 to 85 °C)	RAK3172 datasheet
Currents	Transmit 45 mA at +14 dBm (the datasheet gives only 87 mA at +20 dBm); receive 6 mA for at least 0.1 s or 8 symbols per window; 0.5 s of processing at 3 mA per uplink; sleep 6.0 μ A in all (Section C)	Assumed, to be measured
Ranging cycle	MCU 3 mA for 0.5 s; ultrasonic module 10 mA for 0.4 s (power-up and five pings 60 ms apart); ToF 20 mA for 0.1 s	Assumed typical module figures
Traffic	Two extra uplinks a day for alerts and tip events	Assumed
Cell	Li-SOCl ₂ C (ER26500 class) 8.5 Ah or AA (ER14505 class) 2.6 Ah nominal; 60 % usable for pulses, cold and end-of-life voltage; self-discharge 1 % of nominal a year; 3.67 V open circuit	Typical bobbin-cell data
Radio	11-byte payload plus 13 bytes of LoRaWAN overhead; 125 kHz; coding rate 4/5; 8-symbol preamble; low data rate optimization at SF11 and SF12; US915 slowest rate (DR0, SF10) carries at most 11 bytes of application payload and a 400 ms dwell limit (TTN for the dwell limit; the 11-byte figure is from the LoRaWAN Regional Parameters and was not re-fetched this session)	LoRaWAN defaults, as in FND-CAL-001
Link	+14 dBm; 0 dBi node antenna with 0.5 dB loss; 2 dBi gateway antenna with 2 dB feeder loss, 30 m high; node 1.3 m high; 6 dB noise figure; Okumura-Hata urban model at 868 MHz; container penalty 10 dB (HDPE, antenna under the lid), 30 dB (steel, antenna inside), 3 dB (steel, external lid antenna); 10 dB fade margin	Screening values; container penalties are assumptions to be measured
Lid heat	40 °C ambient; 900 W/m ² sun; absorptance 0.9 (dark lid) or 0.5 (light lid); 25 W/m ² K above and 5 W/m ² K below the lid; unit heat capacity 400 J/K coupled to the lid at 0.75 W/K	Handbook ranges; assumed coupling
Shock	20 g peak when the container strikes the lifter stop; HDPE shear strength 20 MPa; 18 mm washers; M6 stress area 20.1 mm ² at 210 MPa	Assumed; the 20 g figure needs measurement
Mass	ABS 1.05, stainless 7.9, aluminium 2.68 and FR-4 1.85 g/cm ³ applied to model volumes; bought-in parts 160 g (Section I)	Assumed part masses

■ A. Measurement geometry (R1)

- **Depth.** The lid underside is 1,072 mm above the inner floor, and the transducer face sits 1,011 mm above it [A1].

- **Blind zone.** At the adopted 80 % threshold (D6) the waste surface is 202 mm below the face, inside the 250 mm ultrasonic blind zone, so the ultrasonic sensor alone reads fill only up to 75.3 % [A2]. This confirms decision D2.
- **Coverage.** The ToF sensor, used to 400 mm, covers fill from 60.4 % to 100 %, which leaves a 150 mm overlap band where both sensors read and can be compared [A3]. Together they cover the 0 to 1.01 m needed, inside the R1 target of 0.03 to 1.5 m [A4].
- **Beam.** A 15° beam spreads to 0.54 m across at the floor, which averages over uneven waste; the ToF footprint at 0.4 m is 192 mm across [A5]. Mounted at the lid center, the transducer is 492 mm from the nearest wall, so the beam clears the walls down to the floor for half-angles up to 25.9° [A6]. An off-center mount would see echoes from the walls sooner.

■ B. Instrument accuracy (R2)

- **Speed of sound.** Over -20 to 60 °C the speed of sound runs from 318.9 to 365.9 m/s; without compensation the distance error would be -7.1 % to +6.6 % [B1]. The on-board temperature reading removes most of it.
- **Error budget.** With a 5 K error between the sensor and the air column, the residual is 0.85 %, or 8.6 mm at full depth; echo jitter adds 4.3 mm and the timer 0.17 mm [B2]. The ultrasonic instrument error is 9.6 mm and the ToF error 20 mm, against a ± 101 mm allowance (± 10 % of depth), so the instruments use 9 % and 20 % of it [B3].
- **Conclusion.** R2 is decided by the waste surface (bags, cardboard, voids and slopes), not by the sensors. No calculation settles it, so R2 stays at risk.

■ C. Energy and battery life (R4)

Table 2. Sleep current [C1].

Item	Current
Module in stop mode (datasheet 1.69 μ A, rounded up)	2.0 μ A
Accelerometer in wake-on-motion mode (LIS2DH12 class)	2.0 μ A
Temperature sensor in shutdown	0.5 μ A
Nanopower regulator, load switch and divider leakage	0.5 μ A
Buffer capacitor leakage	1.0 μ A
Total	6.0 μA

- **Daily use.** Sleep takes 0.144 mAh a day; each ranging cycle takes 7.50 mAs, or 0.200 mAh a day at 96 cycles; temperature reads every 5 min add 0.0012 mAh a day [C1]. An uplink takes 3.32 μ Ah at SF9 and 19.82 μ Ah at SF12; with two alerts a day the uplinks use 0.086 mAh (SF9, hourly) or 0.278 mAh (SF12, 2-hourly) a day [C2].

Table 3. Daily charge and life on 60 % of nominal capacity [C3].

Case	C cell (8.5 Ah)	AA cell (2.6 Ah)
SF9, hourly	0.664 mAh/day; 21.0 years	0.503 mAh/day; 8.5 years
SF12, every 2 h	0.856 mAh/day; 16.3 years	0.694 mAh/day; 6.2 years

- **R4 is met.** The worst case averages 35.7 μ A including self-discharge and gives 16.3 years on a C cell and 6.2 years on an AA cell, against 5 years minimum and a 10-year design target [C4]. The TRL 2 estimates (about 23 and 18 years) were optimistic by the extra sleep current and the alert traffic, but the conclusion stands: seals and plastics, not the cell, set the service life. The AA cell meets the 5-year minimum but not the 10-year target, which supports decision D3.

■ D. Supply voltage, pulses and heating (R6, R4)

- **Regulator.** A fresh Li-SOCl₂ cell rests at 3.67 V, 70 mV above the RAK3172's 3.6 V maximum [D1]. The carrier therefore needs a nanopower 3.3 V regulator (quiescent current well under 1 μ A, included in Table 2). This is a change from TRL 2, added to BOM item 6 at no change in its indicative price.
- **Pulse load.** Buffering a whole SF12 uplink (45 mA for 1.48 s) in a capacitor with 0.5 V droop would need 0.133 F [D2], and a supercapacitor of that size leaks several microamperes. The design lets the cell carry the pulse, specifies a cell rated for at least 100 mA pulses, and adds a 1,000 μ F low-leakage buffer to cover the voltage delay of a passivated cell. Cold-start voltage delay at -20 °C remains to be tested.
- **Heating is not an option.** A 20 mW heater to keep the ToF window above the dew point would use 133 mAh a day, 156 times the whole budget [D3]. Condensation must be handled by venting, a hydrophobic window and the ultrasonic sensor's tolerance to it.

■ E. Airtime, payload and radio rules (R10, R3)

Table 4. Time on air for an 11-byte uplink (24 bytes on air) at 125 kHz, with two alerts a day [E1].

SF	Per uplink	Hourly (s/day)	Every 2 h (s/day)	EU868 1 % off-time
7	62 ms	1.6	0.9	6 s
8	113 ms	2.9	1.6	11 s
9	206 ms	5.4	2.9	20 s
10	371 ms	9.6	5.2	37 s
11	823 ms	21.4	11.5	82 s
12	1,483 ms	38.6	20.8	147 s

- **Payload size.** The TRL 2 concept used 12 bytes. The slowest US915 rate (DR0, SF10) carries at most 11 bytes of application payload, so a 12-byte payload could not be sent from a US915 sensor at the edge of coverage. The payload is cut to 11 bytes, which fits [E2]; the airtimes are unchanged from TRL 2 because the symbol count does not change. At SF10 an uplink takes 371 ms, inside the 400 ms US915 dwell limit [E3].
- **Fair use.** Hourly reporting at SF12 would use 38.6 s a day, over TTN's 30 s; the 2 h interval at SF11 and SF12 (D6) keeps SF12 at 20.8 s [E4]. SF11 hourly would use only 21.4 s, so the stretch is strictly needed only at SF12 (suggestion for review, not applied).

- **Payload layout.** A candidate 11-byte layout, for the decoder shared with FieldNode (D5): 1 byte version and flags (alert, tip, lid, low battery, sensor used), 1 byte fill in 0.5 % steps, 2 bytes distance in millimeters, 1 byte temperature in degrees Celsius (signed), 1 byte lid-open count, 1 byte tip count, 1 byte battery voltage in 10 mV steps above 2.0 V, 2 bytes minutes since the last emptying and 1 byte spare. This is a sketch for review, not firmware.

■ F. Link budget (R10)

Table 5. Link margin at 1 km and range with a 10 dB fade margin [F1, F2].

Case	SF9 (143.0 dB budget)	SF12 (150.5 dB budget)
HDPE container, antenna under the lid (10 dB)	6.5 dB; 0.80 km	14.0 dB; 1.30 km
Steel container, antenna inside (30 dB)	-13.5 dB; 0.22 km	-6.0 dB; 0.35 km
Steel container, external lid antenna (3 dB)	13.5 dB; 1.26 km	21.0 dB; 2.06 km

- Urban path loss at 1 km is 126.5 dB; gateway sensitivity is -129.5 dBm at SF9 and -137.0 dBm at SF12 [F1].
- **Plastic containers** work to about 0.8 km at SF9 and 1.3 km at SF12 with a 10 dB fade margin. A pilot should place gateways within about 1 km of the containers, or accept SF12 and the 2 h interval.
- **Steel containers with the internal antenna do not meet R10:** even at SF12 the range is only about 0.35 km. An external lid antenna restores 1.3 to 2.1 km (open item O1 in DDR-001).

■ G. Timing (R3, R9)

- **Fill alert.** In the worst case the surface crosses 80 % just after a reading: 15 min to the next reading, up to 147 s of EU868 off-time after a previous SF12 uplink, and 1.5 s of airtime give 17.5 min, inside R3's 20 min [G1].
- **Heat alert.** Read only with the 15 min ranging, a heat alert could take 17.5 min, which misses R9's 15 min. Reading the temperature every 5 min (0.0012 mAh a day [C1]) brings it to 7.5 min [G2]. This is a change from TRL 2.

■ H. Temperature under the lid (R6, R9)

- **Hot lids.** In 900 W/m² sun at 40 °C ambient a dark lid reaches about 67 °C and a light lid about 55 °C [H1]. The unit is bolted to the lid underside and runs close to lid temperature, so on dark lids in hot climates it exceeds R6's 60 °C. The module (to 85 °C) and cell tolerate this; the ultrasonic module's rating must be confirmed.
- **Cold.** The standard RAK3172 is rated to -20 °C, exactly R6's lower limit; the -T variant (-40 °C) gives margin for cold sites.
- **False heat alarms.** The lid's time constant is 5.0 min and the unit's 8.9 min. When 800 W/m² sun breaks through cloud, a dark lid rises 24 K and the unit 15.4 K within 15 min [H2], just over the 15 K rate-of-rise trigger (D7). The rate trigger could therefore fire on a sunny afternoon with no fire. R9 is at risk for false alarms.

■ I. Mass and size (R16)

- **Mass.** Enclosure 74 g, bracket plate and tabs 183 g (1.5 mm stainless, the thinnest in the BOM's range), PCB 17 g and bought-in parts 160 g give 435 g, above R16's 400 g [I1]. With 2.0 or 3.0 mm stainless the unit weighs 496

or 618 g; with a 2.0 mm aluminium (5052 class) bracket it weighs 334 g [I2]. The TRL 2 estimate of 250 g had assumed 70 g for the bracket and bolts. **R16 is not met** with a stainless bracket.

- **Size.** Below the lid the unit measures 150 x 80 x 61.0 mm, inside the 160 x 90 x 100 mm limit [I3].

■ J. Shock and fixing (R7)

- A 20 g shock on the 0.43 kg unit gives 85 N. One washer pulling through the 5 mm HDPE lid needs about 5,655 N and one M6 bolt carries about 4,221 N [J1], so the fixing has a large margin on strength. Loosening under repeated shocks, and IK08 impact, need a test.
- The 50 g cell needs 10 N of retention at 20 g [J2]; the holder is specified with a strap rather than spring clips alone.

■ K. Cost (R14)

- All nine BOM lines are priced; the parts total is \$55.00 against `budget_usd` of \$60, within budget by \$5.00 [K1]. No new budget was recommended, so costs are stated against the one figure.
- An external lid antenna for steel containers (about \$8, open item O1) would take the total to \$63.00, over budget [K2].

■ L. Results against every requirement

Table 6. Requirement status at TRL 3.

ID	Requirement	Target	Value (tag)	Status
R1	Range	0.03 to 1.5 m below the face	0 to 1.01 m needed; ultrasonic 0.25 to 4 m, ToF to 0.4 m (A2 to A4)	Met (datasheet ranges to confirm)
R2	Accuracy on real waste	±10 % of depth for 90 % of readings	Instrument 9.6 mm (ultrasonic), 20 mm (ToF) of 101 mm; surface unknown (B3)	At risk
R3	Reporting and alert timing	Uplink at least every 2 h; alert within 20 min	1 h (2 h at SF11 and SF12); 17.5 min (G1)	Met
R4	Battery life	5 years at SF12; 10-year target	16.3 years (C cell) (C4)	Met
R5	Sealing	IP67 minimum; IP69K where pressure washed	Stock IP67 enclosure	Not met (IP69K)
R6	Climate range	-20 to +60 °C inside the container	Dark lid about 67 °C (H1); module at its -20 °C limit	At risk
R7	Lifting, tipping and lid slams	No loosening; IK08	85 N against 4,221 N per bolt (J1)	Not verifiable at TRL 3
R8	Emptying and lid events	Tip event with the next uplink	Accelerometer wake-on-motion	Met by design
R9	Heat warning	Alert within 15 min above 70 °C or 15 K rise in 15 min	7.5 min with 5 min reads (G2); unit rises 15.4 K in sun (H2)	At risk (false alarms)
R10	Open LoRaWAN within radio rules	Class A; EU868, US915, IN865; duty cycle and 30 s/day	20.8 s/day at SF12 (E4); 11 bytes fits US915 DR0 (E2); steel container 0.35 km (F2)	Not met (steel containers); met for HDPE
R11	Privacy	No camera or microphone; limited payload	BOM and 11-byte layout (E)	Met by design
R12	Installation	10 min or less, four bolts, no cables	Four through-bolts on a 130 x 60 mm pattern	Not verifiable at TRL 3
R13	Tamper resistance	Inside the lid; tamper-resistant fasteners	Button-head tamper bolts from above	Met by design
R14	Cost	\$60 or less	\$55.00 (K1)	Met
R15	Open and interoperable	Documented payload; open decoder	Candidate layout in Section E; decoder shared with FieldNode (D5)	Met by design
R16	Mass and size	400 g or less; within 160 x 90 x 100 mm below the lid	435 g (I1); 150 x 80 x 61.0 mm (I3)	Not met (mass)

Summary: met 8 (R1, R3, R4, R14 by calculation; R8, R11, R13, R15 by design); at risk 3 (R2, R6, R9); not met 3 (R5, R10, R16); not verifiable at TRL 3, 2 (R7, R12).

■ Checks against the TRL 2 figures

TRL 2 claim	TRL 3 value	Change made
Daily use about 0.61 to 0.76 mAh; life about 23 to 18 years (C)	0.664 to 0.856 mAh; 21.0 to 16.3 years	Precis and README updated
AA cell about 10 to 7 years	8.5 to 6.2 years	Precis updated
12-byte payload	11 bytes, so it fits US915 DR0	Precis, requirements and media updated
Uplink airtime about 5 s/day (SF9) and 18 s/day (SF12)	5.4 and 20.8 s/day with two alerts	Precis updated
Mass about 250 g; R16 met	435 g; R16 not met	Requirements, precis and README updated
150 x 80 x 63 mm below the lid	150 x 80 x 61.0 mm (1.5 mm plate)	Requirements and precis updated
Temperature inside a dark lid up to about 60 °C	About 67 °C on a 40 °C day	Problem statement and requirements note updated
80 % fill about 0.21 m below the lid	0.20 m below the transducer face	Precis updated
Parts cost about \$55	\$55.00	None